

Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

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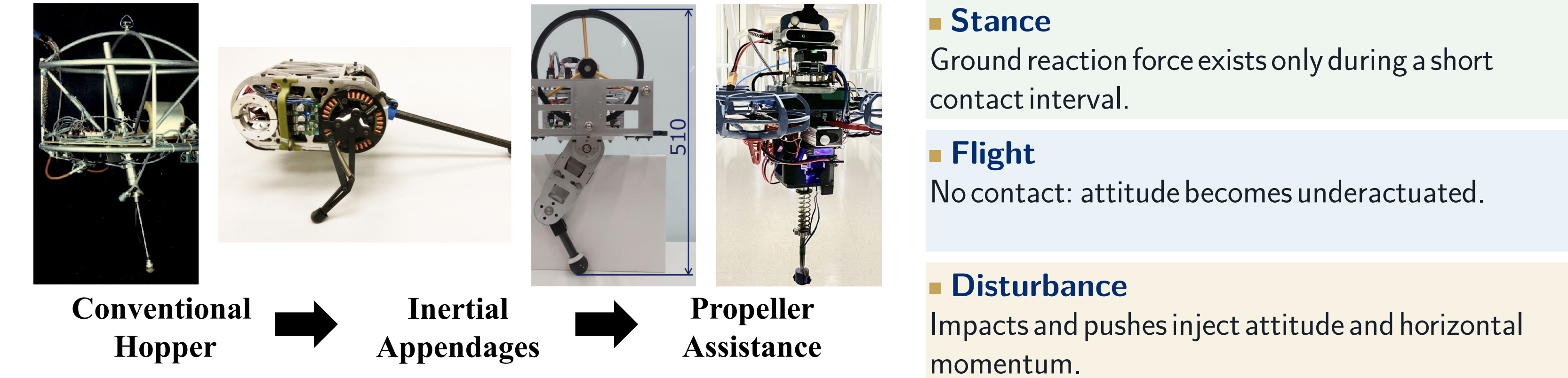


SCAN: DEMO VIDEO
PAPER + ARXIV



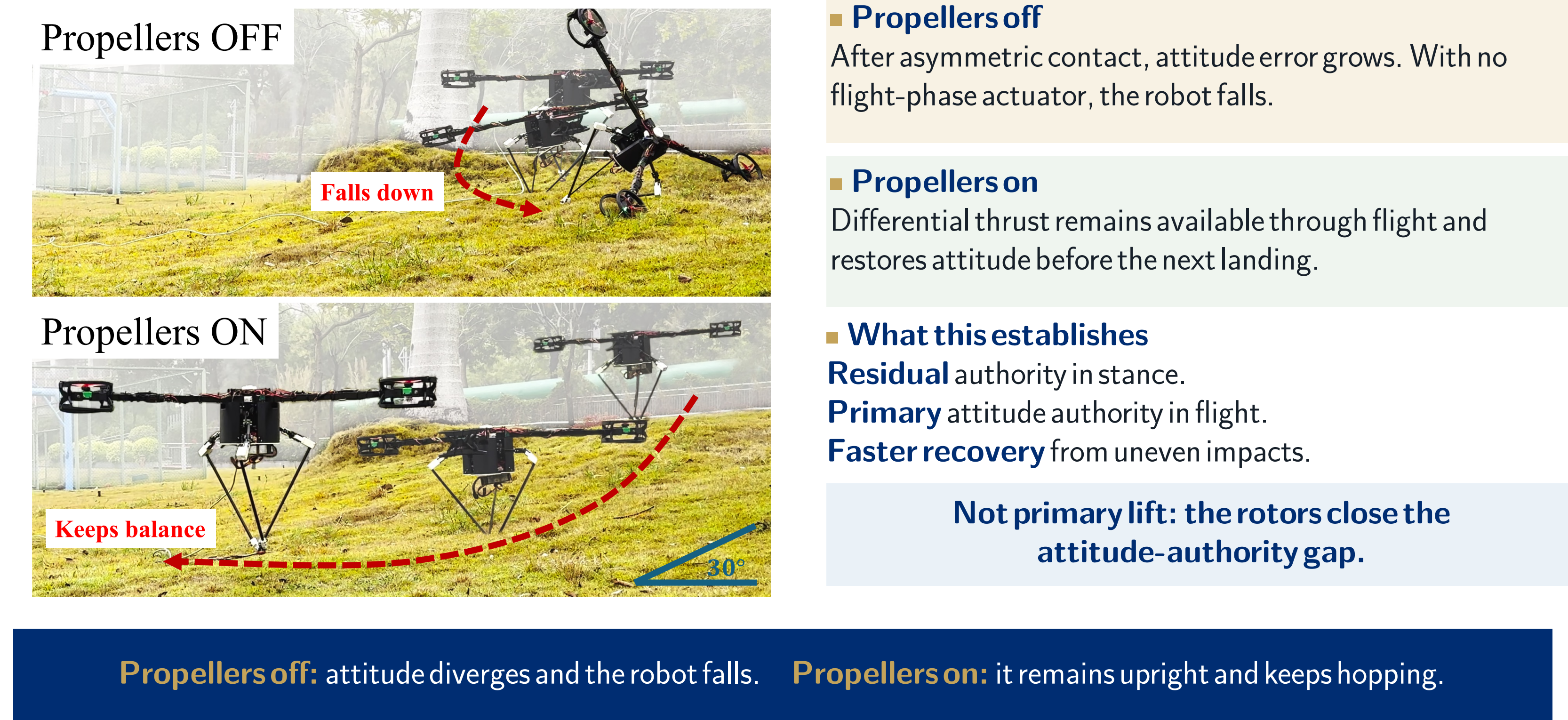
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1 Problem- hybrid dynamics create an authority gap



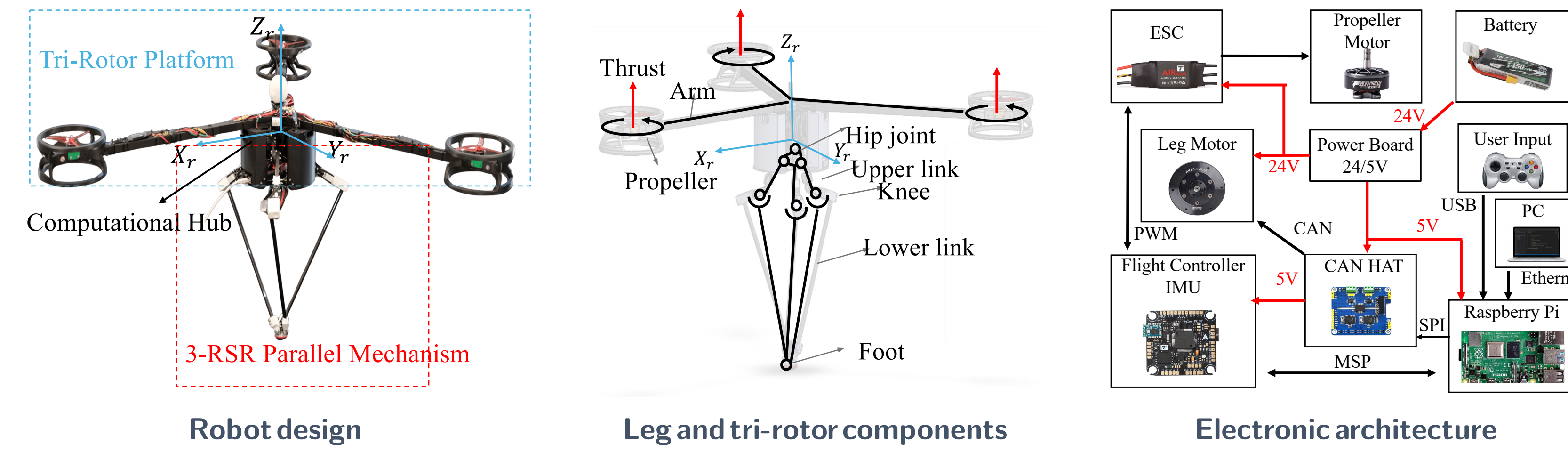
Control authority disappears exactly when attitude can keep drifting.
Design question: how should leg contact force and rotor thrust be coordinated across stance and flight?

2 First proof- stable on 30° grass



Propellers off: attitude diverges and the robot falls. **Propellers on:** it remains upright and keeps hopping.

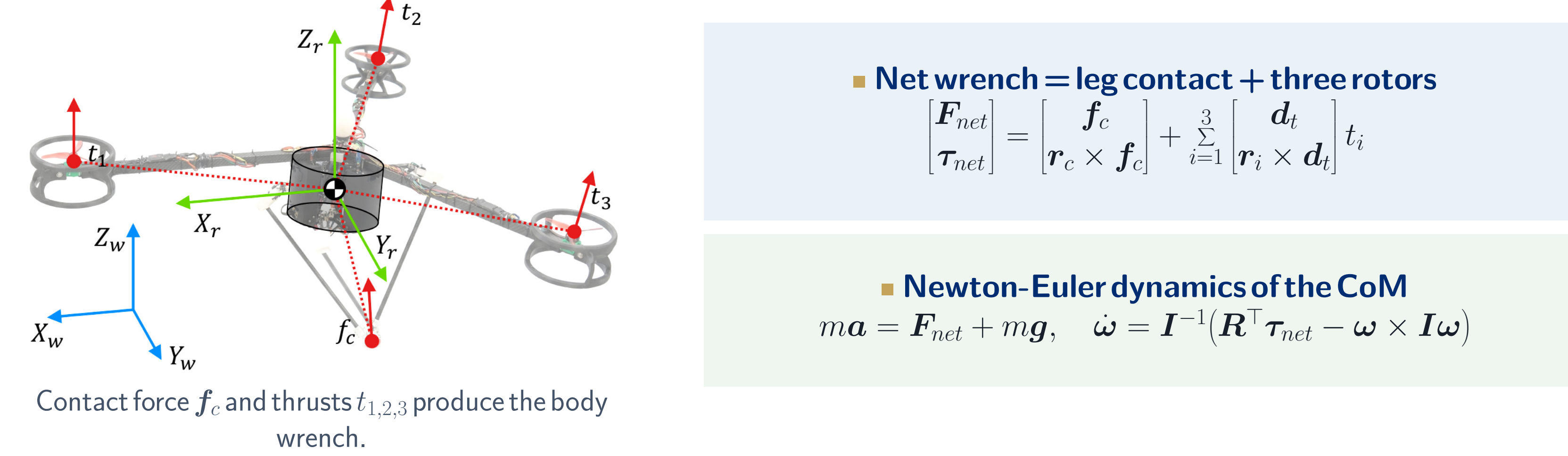
3 Platform- hardware built for leg-first hopping



- 3-RSR parallel leg**
Three RSR chains meet at one foot: fast 3-DoF omnidirectional placement, only **0.35 kg** moving mass.
- Computational hub**
Batteries and electronics near the hip: low center of mass, low inertia.
- Y-layout tri-rotor**
Continuous attitude authority; **10 N** per rotor, total 0.82 mg: a hopper, not a drone.
- External PC @ 500 Hz**
State machine, estimation, wrench planning, and the HFA allocator.
- Raspberry Pi interface**
CAN to the three leg motors; MSP to the flight controller.
- Real-time link**
LCM over Ethernet: RTT ≈ 0.2 ms, 99% < 0.5 ms, within the 2 ms period.

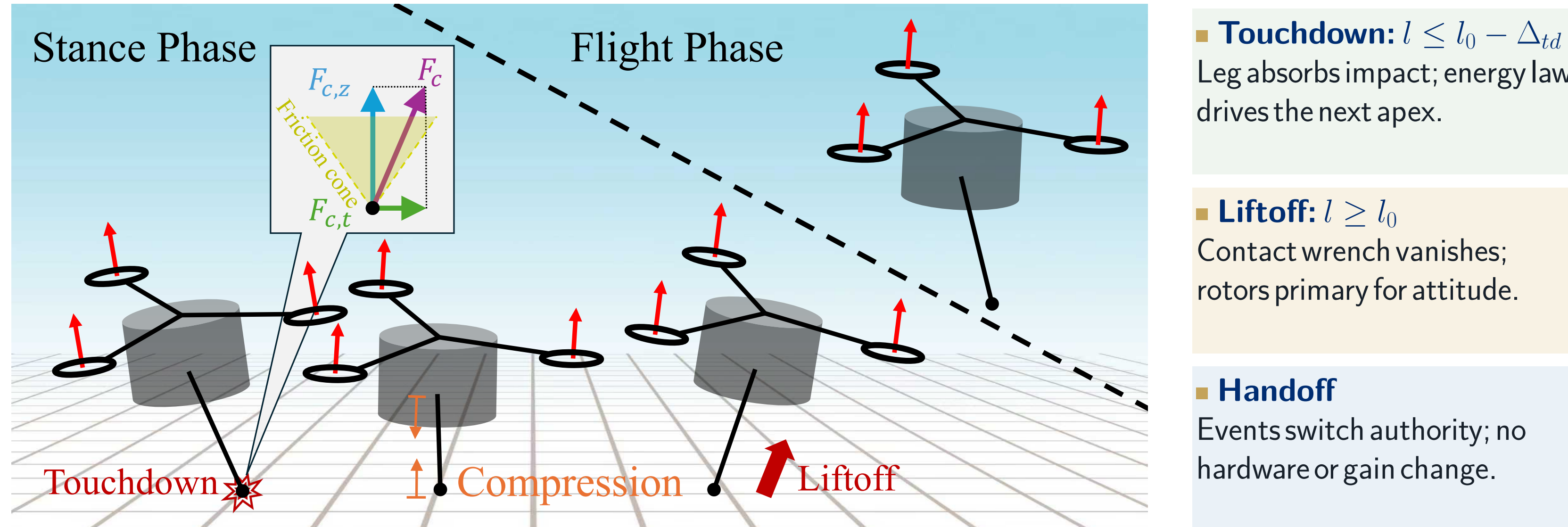
3.75 kg total ■ 9.2% leg mass ■ 20 Nm / leg motor ■ 500 Hz control

4 Model- one rigid body, two actuator sets



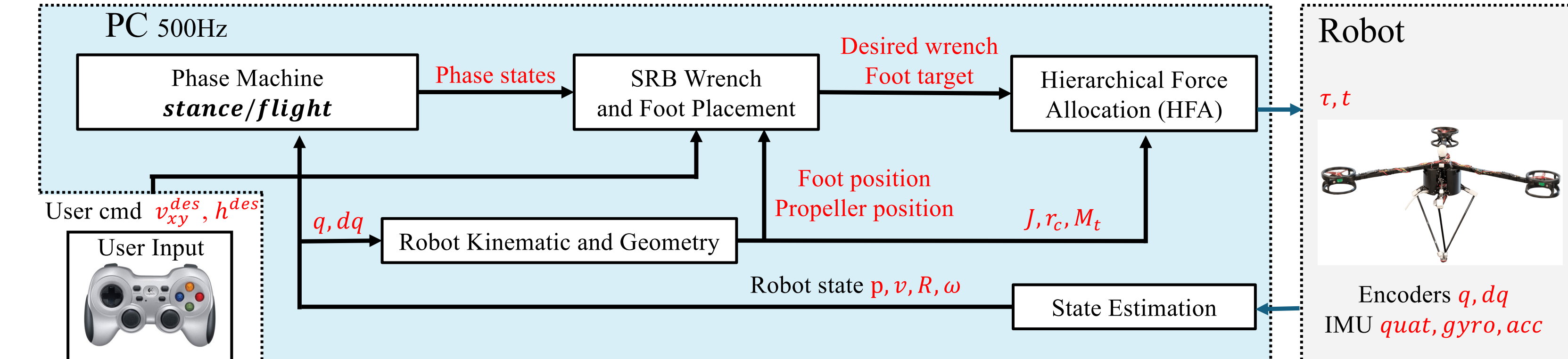
- Symbols (values used in all experiments)**
 f_c : ground reaction force (world) $m=3.75$ kg; g gravity; I inertia
 r_c, r_i : arms: CoM $\rightarrow$ contact, rotor i $R \in SO(3)$ attitude; ω body rate
 $t_i \in [0, 10]$ N thrust along $d_i = Re_3$ J leg Jacobian; $\tau_{max}=20$ Nm limit
 F_{net}, τ_{net} net force, moment $\mu=0.4$ friction coefficient
- $p_z, \dot{p}_z$: CoM height, velocity; $h_{des}=0.664$ m
 l leg length; $l_0=0.464$ m; Δ_{td} threshold
 k_z, b_z, k_E vertical, k_v, k_r Raibert gains
 K_R, K_ω attitude PD gains

5 Onehop- event-driven authority handoff



Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.

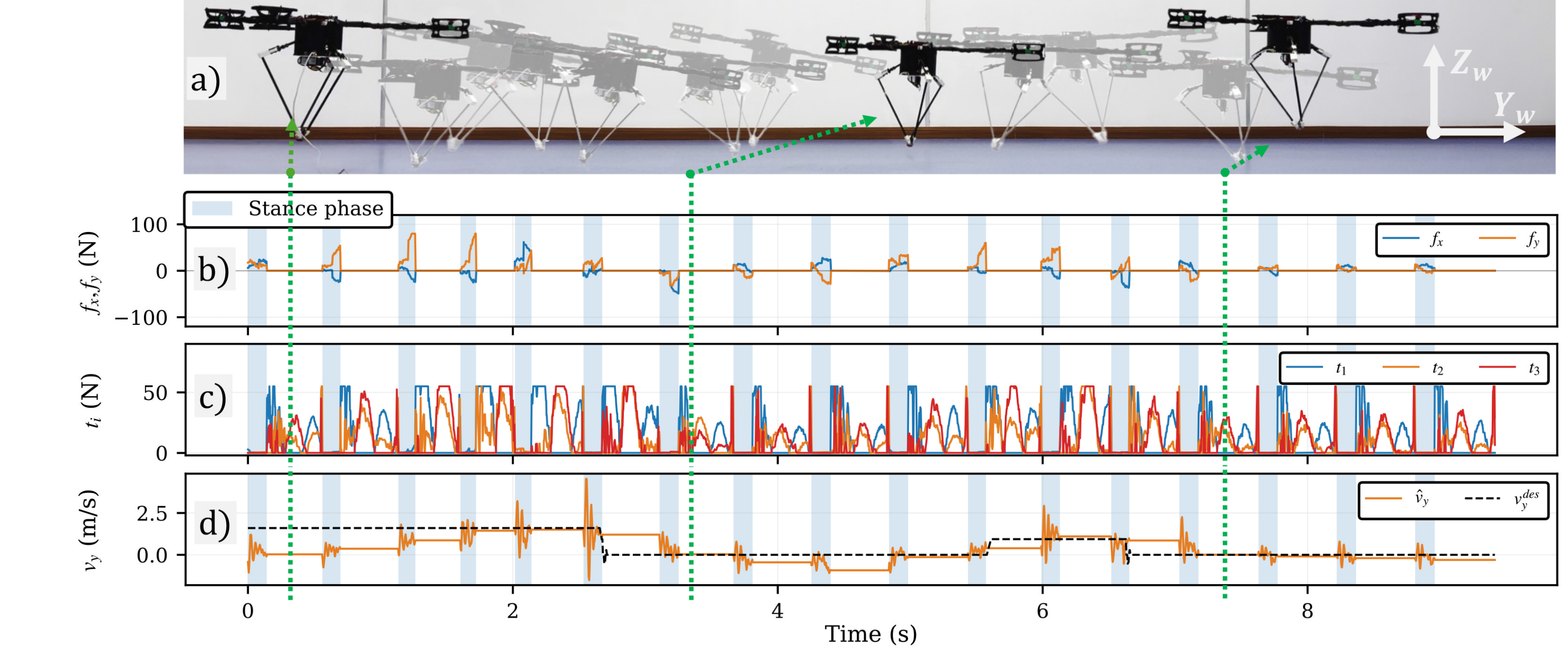
6 Control- leg-priority hierarchical force allocation



500 Hz architecture: commands + feedback $\rightarrow$ planner $\rightarrow$ two-stage allocator $\rightarrow$ leg torques + rotor thrusts.

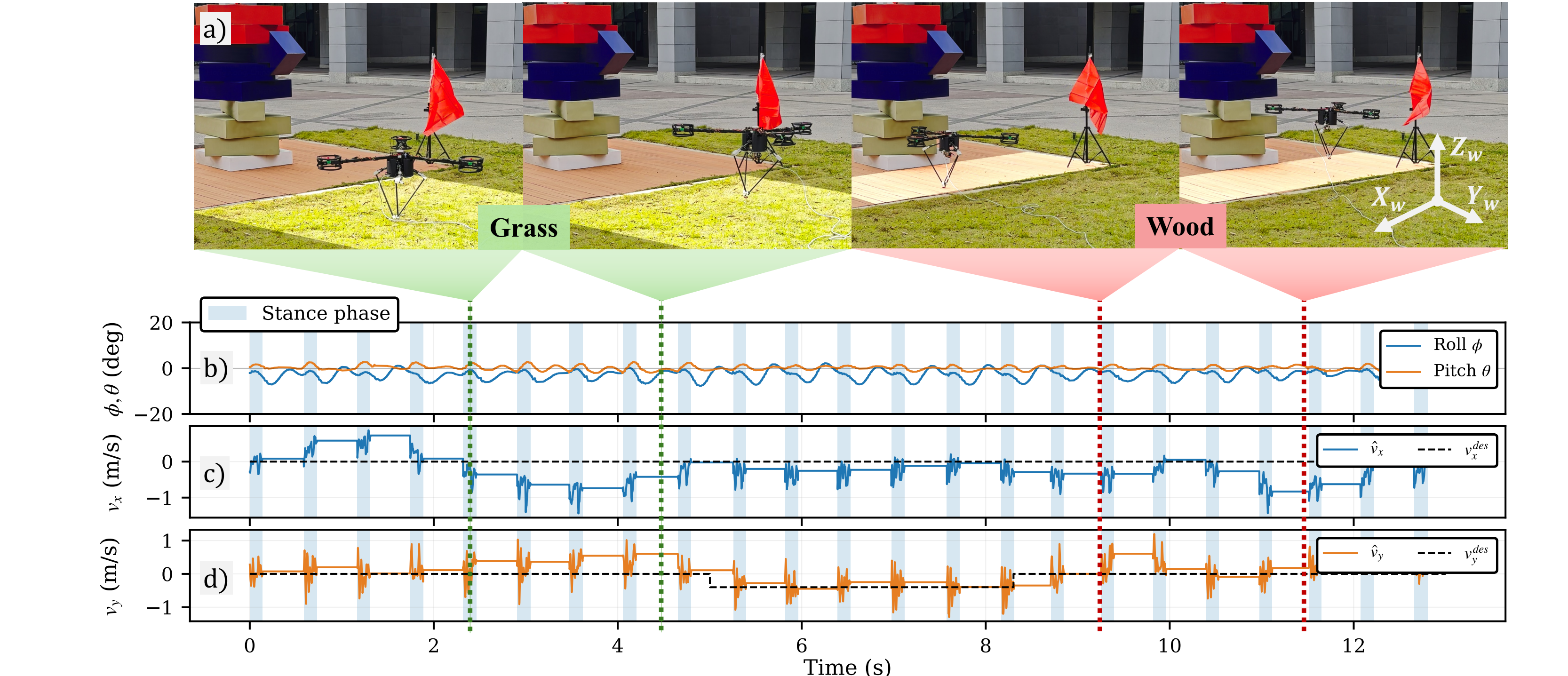
- A. Vertical energy hopping (stance)**
 $E_{sys} = \frac{1}{2}m\dot{p}_z^2 + mgp_z, E_{des} = mgh_{des}$
 $f_{c,z}^{des} = \max(0, k_z(h_{des} - p_z) - b_z\dot{p}_z + k_E(E_{des} - E_{sys}))$
Injects push-off energy, absorbs impact.
- B. SO(3) attitude PD (both phases)**
 $e_R = \frac{1}{2}(R_{des}^T R - R^T R_{des})^V$
 $\tau_b^{des} = -K_R e_R + K_\omega(\omega^{des} - \omega)$
Roll/pitch only; yaw left free. $\tau^{des} = R\tau_b^{des}$.
- C. Leg first- constrained least squares**
 $f_{c,xy}^{cmd} = \arg \min_{f_{xy}} \|(r_c \times f_c)_{rp} - (\tau^{des})_{rp}\|_2^2$
s.t. $\|f_{xy}\|_2 \leq \mu f_{c,z}^{des}, \|J^T(-R^T f_c)\|_\infty \leq \tau_{max}$
The leg realizes all it can under friction and torque limits.
- D. Rotors take only the residual**
 $\tau_{res}^{cmd} = \tau^{des} - r_c \times f_c^{cmd}$
 $\delta t^{cmd} = \arg \min_{\delta t} \|(M_t \delta t)_{rp} - (\tau_{res}^{cmd})_{rp}\|_2^2$
s.t. thrust bounds; $t^{cmd} = \frac{T_{base}}{3} 1_3 + \delta t^{cmd}, T_{base} = 0.02$ mg.
- E. Flight- rotors primary; Raibert-style foot placement**
 $f_c = 0$; rotors track $(\tau^{des})_{rp}$; $\tau_{td,xy}^{des} = k_v v_{xy} - k_r v_{xy}^{des}$
 $p_{foot}^{des} = [x_{td}^{des}, y_{td}^{des}, -\sqrt{l_0^2 - \|r_{td,xy}^{des}\|_2^2}]^T$

7 Validation- indoor stop-and-go tracking



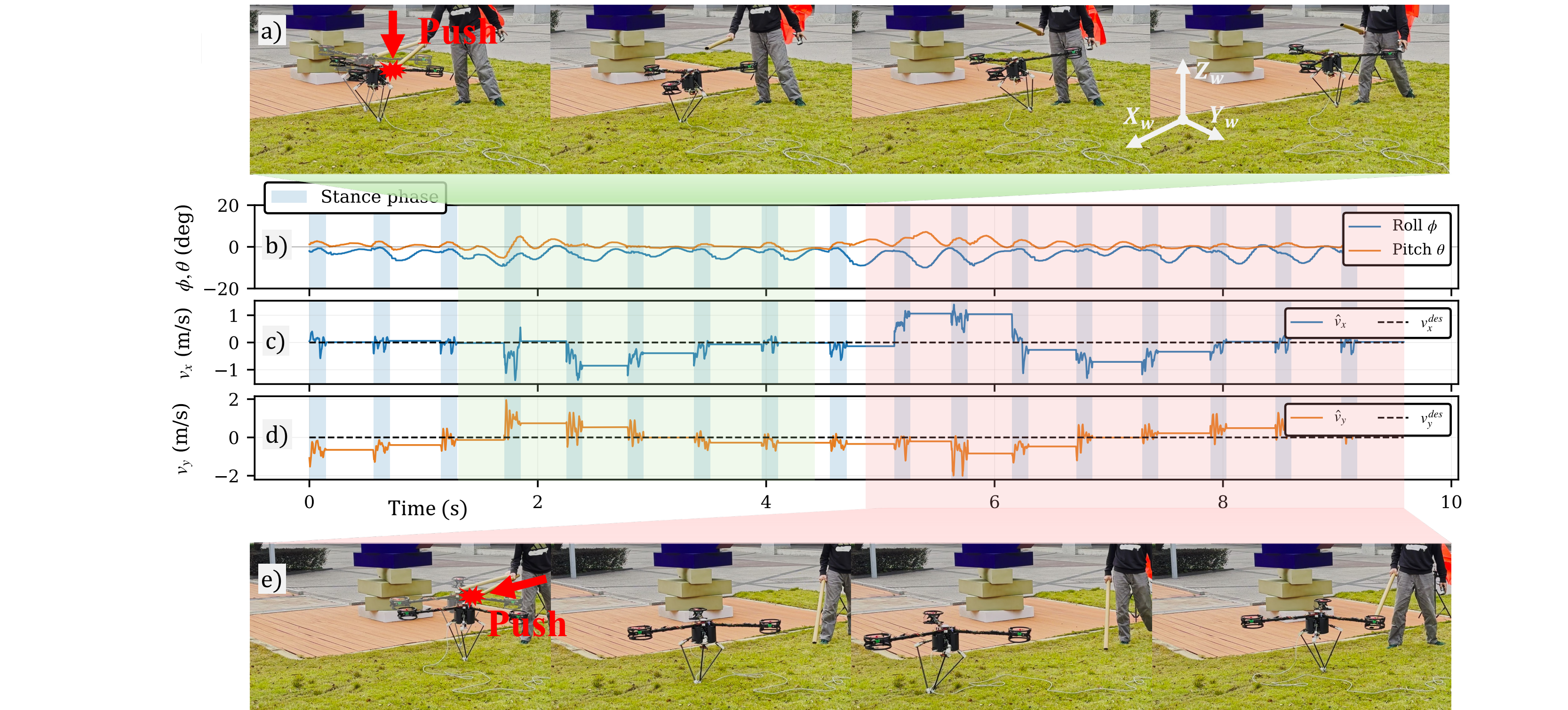
Stop-and-go commands: accelerate, then hard stop (peaks 1.6 and 0.94 m/s) - 17 hops.
Velocity recovered stride to stride by **Raibert-style foot placement** (Panel 6, E).

8 Validation- grass to wood



Grass $\rightarrow$ wood under light wind: **23 hops**- same controller, no terrain model, no gain retuning.
Attitude stays tight: roll $\sigma = 2.1^\circ$, pitch $\sigma = 1.0^\circ$; deviations decay within 1-2 hops.

9 Validation- push recovery



Rotor-arm and body pushes rejected - stable tracking again in **1-3 hops**.